



UNIGLOBAL  
CONSULTANCY

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MASTER ITEM CATALOGUING & AUTO PCODE ENGINE

## Module 1: PCode Generation, Cataloguing & Master Item Engine

Comprehensive 10-week implementation roadmap, universal interchange matrix, legacy SKU cleansing (50,000+ items), dedicated engineering resource allocation, and milestone-governed payment structure for **Popular Auto Spare & A/C Parts Co. W.L.L.**

TOTAL MODULE FEE

**BD 3,409.091**

IMPLEMENTATION TIMELINE

**10 Working Weeks**

DELIVERY EFFORT

**2.50 Engineering Months**

MILESTONE TRANCHES

**4 Verified Gates (25%)**



**50,000+**

Legacy SKUs Cleansed



**< 100ms**

Search Response Latency



**1-to-Many**

OEM Interchange Cross-Ref



**15-Day UAT**

Verification Grace Period

# Functional Scope & Technical Architecture Breakdown

Module 1 establishes the bedrock of the entire Popular ERP ecosystem by introducing deterministic, algorithmically generated unique Popular Codes (PCode) alongside a centralized item master with multi-brand hierarchy and OEM cross-referencing.

## Smart PCode Generation Engine

Algorithmic generator producing unique alphanumeric tokens based on Category, Sub-Category, Make, Model, and Part Family, eliminating duplicate item creation across branches.

## Multi-Brand & Vehicle Hierarchy

Structured relational taxonomy supporting Japanese, Korean, American, and European vehicle brands, model years, chassis codes, and engine specifications.

## Universal Interchange Matrix

1-to-Many and Many-to-Many cross-referencing between Genuine OEM part numbers, Aftermarket brand codes (Denso, Bosch, Valeo), and Popular internal PCodes.

## Fast Multi-Param Fuzzy Search

Full-text indexing with soundex and typo-tolerant search across Arabic/English part names, partial part numbers, OEM codes, and car models with <100ms response.

# Dedicated Engineering Team & Resource Rate Matrix (BHD)

Pricing for Module 1 is determined directly by allocating the 5-person dedicated engineering team over 10 working weeks (2.50 calendar months):

**Project Manager & Solutions Architect**

**BD 545.455**

BD 136.364 / Wk • 2.50 Mo  
= BD 1,363.638

100% Dedicated

**Back-End Lead Engineer (PHP MVC)**

**BD 227.273**

BD 56.818 / Wk • 2.50 Mo  
= BD 568.183

100% Dedicated

**Front-End Lead Engineer (React 18)**

**BD 227.273**

BD 56.818 / Wk • 2.50 Mo  
= BD 568.183

100% Dedicated

**Database & Cloud DevOps Specialist**

**BD 204.545**

BD 51.136 / Wk • 2.50 Mo  
= BD 511.363

100% Dedicated

**QA Automation & Multi-Branch Test Lead**

**BD 159.091**

BD 39.773 / Wk • 2.50 Mo  
= BD 397.728

100% Dedicated

| ENGINEERING DESIGNATION                          | MONTHLY RATE (BHD) | WEEKLY RATE (BHD) | HOURLY RATE (160H) | EFFORT ALLOCATION    | TOTAL COST (BHD) |
|--------------------------------------------------|--------------------|-------------------|--------------------|----------------------|------------------|
| Project Manager & Enterprise Solutions Architect | BD 545.455         | BD 136.364        | BD 3.409           | 2.50 Months (10 Wks) | BD 1,363.638     |
| Back-End Lead Engineer (PHP MVC / REST APIs)     | BD 227.273         | BD 56.818         | BD 1.420           | 2.50 Months (10 Wks) | BD 568.183       |
| TOTAL MODULE 1 DEDICATED ENGINEERING INVESTMENT  |                    |                   |                    | 10 Working Weeks     | BD 3,409.091     |

| ENGINEERING DESIGNATION                           | MONTHLY RATE (BHD) | WEEKLY RATE (BHD) | HOURLY RATE (160H) | EFFORT ALLOCATION    | TOTAL COST (BHD) |
|---------------------------------------------------|--------------------|-------------------|--------------------|----------------------|------------------|
| Front-End Lead Engineer (Admin UI / React Tokens) | BD 227.273         | BD 56.818         | BD 1.420           | 2.50 Months (10 Wks) | BD 568.183       |
| Database & Cloud Infrastructure Architect         | BD 204.545         | BD 51.136         | BD 1.278           | 2.50 Months (10 Wks) | BD 511.363       |
| QA & System Test Automation Lead Engineer         | BD 159.091         | BD 39.773         | BD 0.994           | 2.50 Months (10 Wks) | BD 397.728       |
| TOTAL MODULE 1 DEDICATED ENGINEERING INVESTMENT   |                    |                   |                    | 10 Working Weeks     | BD 3,409.091     |

## Milestone Delivery Breakdown & Payment Tranches

Module 1 is invoiced strictly across 4 equal milestone delivery gates upon verified demonstration and UAT staging sign-off:

| Milestone Phase               | Deliverable Description & Scope                                             | Timeline   | Share (%) | Invoice Amount (BHD) | Acceptance Verification Trigger                                        |
|-------------------------------|-----------------------------------------------------------------------------|------------|-----------|----------------------|------------------------------------------------------------------------|
| Milestone 1                   | PCode Database Schema, Legacy Data Cleansing Scripts & Base Token Generator | Weeks 1–3  | 25.00%    | BD 852.273           | Live demonstration of cleansed schema & algorithmic PCode generator    |
| Milestone 2                   | Brand / Model / Spec Hierarchy & Automated Dynamic Category Code Engine     | Weeks 4–5  | 25.00%    | BD 852.273           | Sign-off of Make/Model vehicle hierarchy and dynamic categorization    |
| Milestone 3                   | Universal Cross-Reference / Interchange Matrix & 1D/2D Barcode Tagging      | Weeks 6–8  | 25.00%    | BD 852.273           | Validation of OEM-to-Aftermarket interchange matrix and barcode output |
| Milestone 4                   | Fast Search Engine (<100ms), 50,000+ SKU Data Migration & Final Branch UAT  | Weeks 9–10 | 25.00%    | BD 852.273           | Final production migration sign-off and multi-branch UAT acceptance    |
| TOTAL MODULE 1 CONTRACT VALUE |                                                                             | 10 Weeks   | 100.00%   | BD 3,409.091         |                                                                        |

## Terms, SLA & Governance Protocols

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### 1. 15-Day UAT Review SLA

- Popular Auto Spare is allocated a 15-calendar-day UAT review window per milestone to test all acceptance criteria.
- Any bugs discovered during UAT are remediated at zero additional cost within 48 business hours.

### 2. Source Code & IP Transfer

- Full proprietary ownership of PCode generation algorithms, schemas, and indexing routines transfers to Popular Auto Spare Co.
- Zero ongoing license fees or third-party proprietary royalties.

## Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures and authorizations below, all participating executive and technical stakeholders ratify the deliverables, implementation timeline, SLA commitments, and financial disbursement gates for **Module 1: PCode Generation & Item Master Engine (SL-POP-ERP-MS-001)**:

### 1. Engineering Preparation & Stakeholder Verification Matrix

|                                                                                                                                                       |                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DEVELOPER / ARCHITECT</b><br><b>Ancy Varghese Thekkan</b><br>Software Architect, SaNDS Lab<br>✓ PREPARED (10-SEPTEMBER-2026)<br>COMPLETED          | <b>VERIFIED BY (DEVELOPER)</b><br><b>Ajit Kumar KV</b><br>CEO & Managing Director, SaNDS Lab<br>✓ VERIFIED (21-SEPTEMBER-2026)<br>CHECKED & VERIFIED        |
| <b>CONSULTANT REVIEW</b><br><b>UniGlobal Business Solutions</b><br>Project Management Consultant<br>✓ REVIEWED (15-MARCH-2026)<br>REVIEWED & ENDORSED | <b>CLIENT APPROVAL</b><br><b>Popular Auto Spare &amp; A/C Parts</b><br>Authorized Executive Signatory<br>✓ APPROVED (22-FEBRUARY-2026)<br>APPROVED & SIGNED |

### 2. Executive Multi-Party Digital Acceptance Console

|                                                                                                                  |                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Popular Auto Spare Co. W.L.L</b><br>Client Executive Sign-off & Commercial Approval<br>No Signature on Record | <b>SaNDS Lab Middle East W.L.L</b><br>Lead Solutions Architect & Engineering Director<br>No Signature on Record<br>PENDING SIGN-OFF |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

PENDING SIGN-OFF

## UniGlobal Consultancy

Executive Strategic Advisor & Governance Lead

 No Signature on Record

PENDING SIGN-OFF

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Popular Auto Spare & A/C Parts Co. W.L.L • Kingdom of Bahrain • Commercial Registration (CR) Active

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